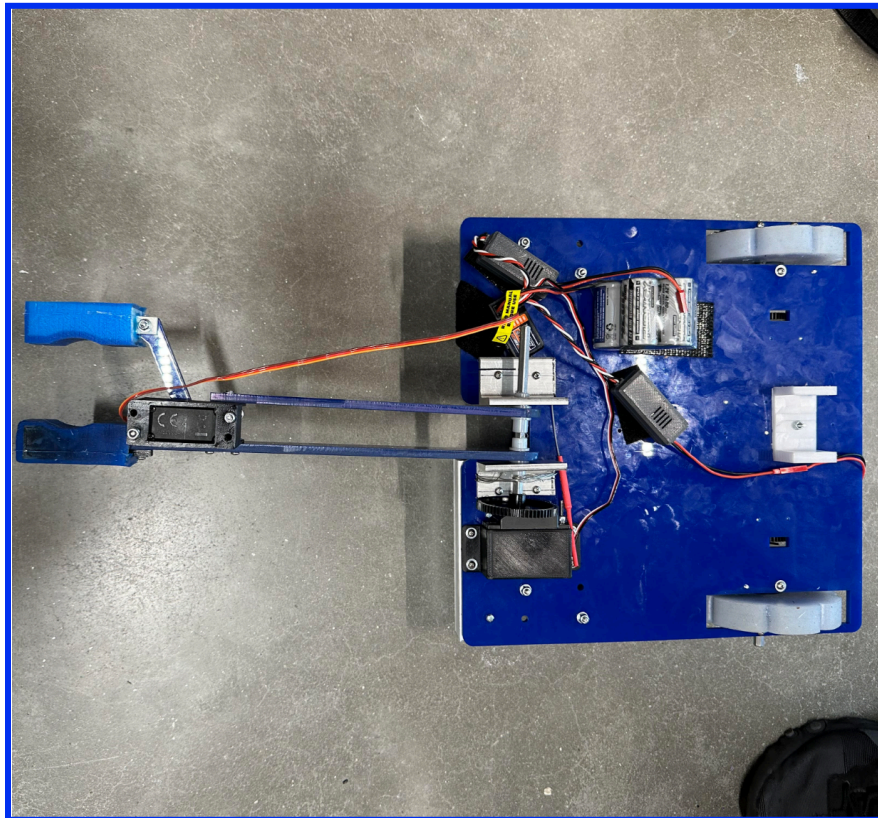


THE BLUE TEAM

ES-51 “Turf Wars” Final Report

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ABSTRACT

For the Spring 2023 Turf Wars competition each team was tasked with the following: design and build a remote controlled robot to overcome obstacles and complete tasks within the competition arena. Specifically, the robot would have to interact with wooden blocks, hoops and spheres and place them in contact with a goal-structure to reap points. This report details the engineering and design process of “The Blue Team” (Team 7). While The Blue Team’s process was not linear, and involved multiple changes of course and failed ideas, the group ultimately created a successful robot fulfilling the requirements put forth by the Turf Wars competition. The final design is a robot with two-wheel drive integrated into its custom acrylic chassis. The mechanism to interact with the blocks is a “grabber-arm” based on an extending arm base and closing pinchers, powered by two servo motors.

CONCEPT DEVELOPMENT

Competition Constraints

- The Box of Justice is a 12” x 12” x 12” hollow cube into which your robot must fit
- Each team will be allowed 2 minutes for combined set-up and take-down of their robot.
- The final tournament during the reading period, double elimination tournament.
- Every round of the tournament is 4 minutes long.
- The highest scoring robot in each round will advance.
- Intentional contact with an opponent’s robot is not allowed
- Robot Death is defined as an inability to progress forward (e.g. flipped over, stuck, can’t make it over an obstacle, etc.) Lives are used to recover from Robot Death. Teams are allowed 5 lives for the entire tournament.

Building Constraints:

- Your robot must be made from the kit materials from a specified list of approved items
- Each team must use each of the methods learned in class for a critical component of their robot. Milling, laser cutting, turning (lathe), 3D printing, drilling, and molding.
- Acrylic is limited to the amount provided at the start of the semester.
- No liquids, chemicals, welding, or compressed gas are permitted.
- Cable (zip) ties are only allowed for cable management.
- Glue and tape are not permitted to be used on the robot
- 15 cubic inches of 3D printing material is allowed (roughly equivalent to \$70 in materials).
- Teams must use at least 2 (max 3) of the electric screwdriver motors given to your team.
- The current draw of your motors may not exceed 9A.
- Only the motors and servos provided by the teaching staff may be used.
- Using sandpaper or nails for traction is not allowed
- Anything that damages the arena is forbidden.
- Belts may be used to transmit power but may not be used as tank tracks.
- No dangerous machines are allowed.

Criteria

Table 1. Pugh Matrix

We decided that all of the criteria would be weighted equally

Criteria	The Scooper	The Crane	The Forklift	Grabber (Final Design)
Size	3	3	3	3
Weight	3	3	3	3
c.o.g	4	2	3	5
Speed	3	2	4	4
Scoring	2	5	4	4
Simplicity	3	1	4	5

Alternative Solutions:

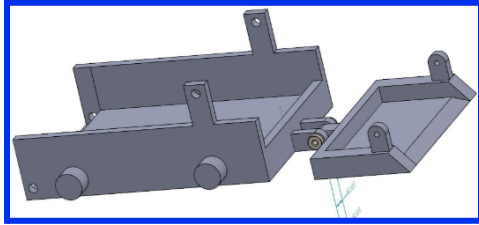


Figure 1. The Scooper

Figures 1-3 show our first three alternative solutions. We presented these ideas in the first team design review. The scooper (Fig.1) was designed to collect blocks in its “bed” with its scoop mechanism. It was designed to collect enough blocks to prevent the other team from being able to score points. Ultimately, we elected not to use the scooper because it would have difficulty lifting the blocks to the upper levels of the goal post.

Figure two was the original crane design. We chose not to continue with the crane design for a number of reasons. Primarily, the design seemed too complex to build. Having multiple pivot points in the arm would be helpful if the task required extreme precision. However, the task of stacking blocks on the goal can be achieved just as well with only one joint pivoting an arm. As it would turn out, our final design would come to resemble the crane with a simplified arm design.

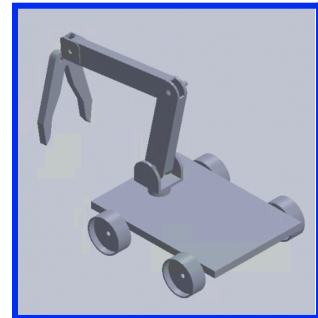


Figure 2. The Crane →

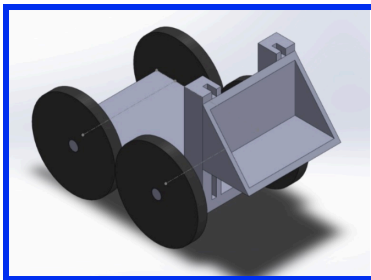


Figure 3. The Forklift

By the end of design review 1, we had decided that The Forklift would be our final design. It appeared to be simpler than the previous two designs, and would require fewer additional servo motors. We ultimately thought that this design would be the best at scoring, and should thus become our final design. However, before design review 2, our team had a change of heart. We realized that The Forklift would ← have trouble depositing blocks onto the goal post once they’d been picked up, and introducing linear motion to the mechanism could potentially cause reliability issues.

By the second design review, we had realized that we would need to change direction. We thought it would be simple and very effective to simply use a “grabber” design inspired by the yellow longarm in the makerspace. Our original idea included a pole at the rear of the robot which would use a winch mechanism to reel up the arm. We included another servo at the base of the arm which was supposed to control the claw mechanism with a belt and pulley system running along the length of the arm.

Figure 4. Grabber Schematic

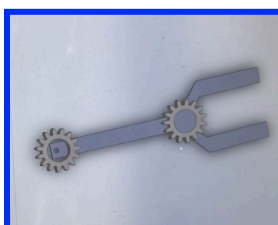
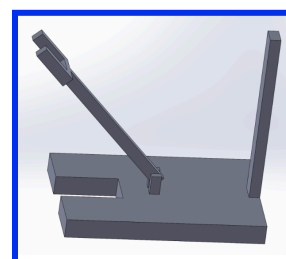


Figure 5. Grabber design



ANALYSIS

Experiments



In order to diagnose issues we had with our arm mechanism we attempted picking up different objects with varying gear meshes to determine if our issues resided in the GR or the shaft instability. We determined we needed both a higher GR for the arm and a tighter clearance for our shaft supports.

← **Figure 6. Arm Mechanism Trials**

Calculations

$$T = (mgr)/n * (\sin\theta + v/gt)$$

Equation 1: Wheel Torque Calculation

Where m = mass of robot, g = acceleration due to gravity, n = number of wheels, v = velocity, t = time, theta = ramp angle

$$((3m/s) * (9.81m/s^2) * (.04445m))/2 * (\sin(30) + (0.15m/s)/(1s) * (9.81m/s^2)) = .337 Nm$$

$$T = F * d$$

Equation 2: Arm Torque Calculation

Where F = force, d = distance, T = Torque

Total mass of arm and load (considered as a point mass at end of arm) = .283 kg

We rounded the above number up to .375 kg to ensure the servo could lift the arm.

$$25cm * .375kg = 9.375 kg * cm$$

Torque from a continuous servo motor: 5.1 kg-cm

Thus, we elected to use a continuous servo with a gear setup with a gear ratio ≥ 2 . We chose a GR of 4 to ensure we were well in range of the necessary torque.

FINAL SOLUTION

Description of Functionality

Our final design implements the grabber arm described in the alternative solutions section with a number of improvements. We decided to simply place a servo motor at the base of the arm with a gear ratio large enough to lift loads at the end. We originally had a gear ratio of 2, as our calculations demonstrated, but we found that in practice it was better to have a larger gear ratio, as our arm sometimes struggled to lift the blocks.

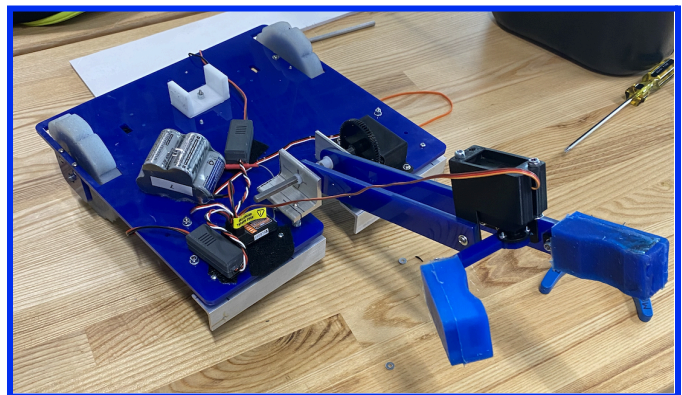


Figure 7. Complete Photo of Robot

The second servo is placed at the end of the arm directly onto the claw mechanism. This requires no gear ratio. We made one of the claws stationary, and the other able to pivot. Both claws are coated in silicon for improved friction and grip.

Advantages, How does it satisfy criteria, How did it perform?

Our robot performs really well, and satisfies the criteria included in our Pugh matrix. It's a simple design which made it much easier to build and also thus has fewer factors that could possibly lead to failure. It is really efficient at scoring, and consistently places blocks and spheres on the top shelf. The center of gravity of the design appears to be close to the middle of the robot. While the gear boxes we designed through the course of the class are in the back, the battery and the arm are placed in the front, evening out the weight distribution. The majority of the mass is low to the ground so it is unlikely to flip over. While the arm reaches 25cm beyond the front of our robot, we used a continuous servo motor, allowing it to flip back on itself, and thus remain inside the box of justice. Our robot performed well during Turf Wars and did not require any repairs or adjustments in between the rounds, and we did not have to use any of our lives, which highlights the reliability of our simple design. One could argue that the reliability of our robot comes close to that of a Toyota Tacoma. However, it was tricky to line up the arm with objects quickly to pick them up, and it required lots of practice to be efficient.

Disadvantages, where could it be improved

There are a number of ways in which our robot could be improved. Our robot is only capable of moving one block at a time. This makes the process of point scoring time consuming. Other teams designed their robots to pick up multiple blocks at one time, which might've been a more effective method. Also, our robot struggles to pick up hoops. While there will be more than enough blocks and spheres on the table to score many points, we would struggle to get points from hoops. Another issue we ran into with our robot is that often the gears on the motos become unmeshed. We tried to resolve this problem by putting on plastic washers, but we were not sure that this would prevent the gears from failing during the competition.

FINAL DESIGN SPECIFICATIONS

Mass	Dimensions	Undercarriage Clearance	Turning Radius	Drivetrain GR	Arm GR	Top speed
6.8 lb	11.5 in. x 11.5 in.	0.78 in	10 in	3	4	0.9 mph

APPENDIX

Bill of Materials

ITEM NO.	PART NUMBER	DESCRIPTION	MANUFACTURING TECHNIQUE	MATERIAL	QTY
1	Custom Chassis	11.5" x 11.5" Base w/ Cutout for Arm	Designed in SW and lasercut	Acrylic	1
2	Wheel Mount	Chassis Side Mounts for Wheel Stability	Designed in SW and lasercut	Acrylic	2
3	2358T51	Static-Control Plate Caster Wheels		Chrome ; Polypropylene	2
4	CR-PHMS 0.19-24x0.4375x0.4375-N	Caster Plate Screws		Steel	8
5	MSHXNUT 0.190-24-S-N	Hex Nut		Steel	8
6	Motor	Motor from screwdriver		Misc.	2
7	Gearbox Motor Mount	Motor Mount	Designed in SW and 3D printed	3D Print Filament	2
8	91772A110	Pan Head Machine Screw, 4-40 Thread, 0.5" Length		Steel	8
9	92010A022	M2.5 x 12 mm Flat Head Screw		Steel	4
10	90126A505 Washer	Washer for Number 4 Screw Size, 0.125" ID, 0.312" OD		Steel	25
11	91772A116	Pan Head Machine Screw 4-40 Thread 1.25"		Steel	8
12	94639A662	Unthreaded Spacers, 3/16" OD, 11/16" Length		Plastic	8
13	Planetary Gearbox	Planetary Gearbox from Screwdriver		Iron	2
14	Lasercut Front Plate	Lasercut Front Plate	Lasercut given design template	Acrylic	2
15	Metal Gear-Shaft Interface	Metal Circle with 6 Notches from Screwdriver		Iron	2
16	Hex Drive Shaft	Hex Drive Shaft from Screwdriver		Steel	2
17	Washer Screwdriver Large	Washer 0.39" ID, 0.77" OD from Screwdriver		Steel	2
18	Delrin Support	Delrin Shaft Support	CNC milled w/ given dimensions	Delrin	2
19	Hex Shaft	Hex Shaft		Steel	2
20	Washer Screwdriver Small	Washer 0.41" ID, 0.54" OD from Screwdriver		Steel	2
21	98541A118	External Retaining Ring for 10mm Shaft Diameter		Steel	2
22	Gear 16T 32DP Broached 025	Acetal Plastic Gear, 16 Teeth, 32 Pitch		Plastic	2

23	Plastic Sleeve Bearing Broached	Nylon Flange Bushing, ID 1/4" OD 3/8"		Nylon	2
24	97431A300	E-Clip 0.25" ID		Steel	2
25	95606A120	Nylon Washer 0.25" ID		Nylon	2
26	Gear 48T 32DP Drilled Spring Pin 00625	Acetal Plastic Gear, 48 Teeth, 32 Pitch		Plastic	2
27	92373A113	Slotted Spring Pin, 1/16" Diameter, 3/4" Long		Steel	2
28	Aluminum Shaft Support	Aluminum Shaft Support		Aluminum	2
29	91772A108	Pan Head Machine Screw, 4-40 Thread, 3/8" Length		Steel	28
30	Plastic Sleeve Bearing	Nylon Flange Bushing, ID 1/4" OD 3/8"		Nylon	2
31	Hex Shaft	Hex Shaft in Gearbox Driving Wheels		Steel	2
32	Custom Treaded Wheels	Elliptically-Treaded Silicon Wheels	Mold for wheels CNC milled and filled with acrylic insert and molded w/ Ecoflex	Acrylic; Ecoflex 00-50	2
33	180 Rotation Servo	180 Degree Rotation Servo		3D Print Filament	1
34	Spring RC Continuous Servo	Continuous Rotation Servo		Misc.	1
35	Aluminum Shaft Support; One Hole; CUT	Aluminum Shaft Support	Aluminum bracket drilled and trimmed with vertical bandsaw		1
36	Plastic Sleeve Bearing Hex	Nylon Flange Bushing, ID 1/4" OD 3/8"		Plastic	4
37	Aluminum Shaft Support One Hole	Aluminum Shaft Support	Aluminum bracket drilled	Aluminum	1
38	Arm Slab Short	Acrylic Arm Slab for Moving End	Designed in SW and lasercut	Acrylic	1
39	Arm Slab Long	Acrylic Arm Slab for Stationary End	Designed in SW and lasercut	Acrylic	1
40	Hex Shaft 5.5 In	5.5" Cut Hex Shaft for Arm Mechanism		Steel	1
41	Servo Round Attachment	Servo Attachment Joining Servo and Pincher-Attachment		Plastic	1
42	Power HD Standard 6001HB Servo	180 Degree Rotation Servo		Misc.	1
43	Stationary Pincher Insert	3D Printed Insert to be Molded	Designed in SW and 3D Printed w/ 1.93 cubic in filament	3D Print Filament	1
44	Moving Pincher Insert	3D Printed Insert to be Molded and joined to Servo	Designed in SW and 3D Printed w/ 1.93 cubic in	3D Print Filament	1
45	Pincher Servo Mount	Servo Mount Screwed onto Acrylic Arm Slabs	Designed in SW and 3D Printed w/ 1.86 cubic in	3D Print Filament	1
46	Pincher-Servo Attachment	Attachment Joining Servo Attachment and Moving Pincher Arm	Designed in SW and lasercut	Acrylic	1

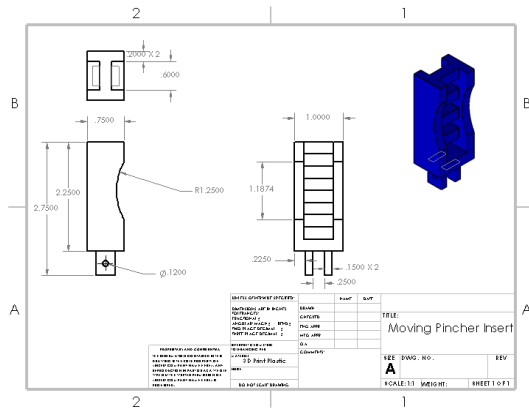
47	Arm Servo Mount	Servo Mount Moving Arm Base	Designed in Sw and 3D Printed w/ 1.44 cubic in	3D Print Filament	1
48	Gear 64T 32DP	64 Teeth Plastic Gear		Plastic	1
49	Servo Gear Attachment	Servo Gear Attachment		Plastic	1
50	SBHCSCREW 0.073-72x0.125-HX-N	Panhead Screw		Steel	1
51	SL-PHMS-MINI 0.0217-203x0.1x0.1-N	Panhead Miniature Screw		Steel	4
52	Arm Stopper	U-Shaped Arm Stopper Holding Arm Base	Manually CNC milled	Delrin	1
53	Barrier Right	Aluminum Shaft Support		Aluminum	1
54	Barrier Left	Aluminum Shaft Support		Aluminum	1

Required Fabrication Methods

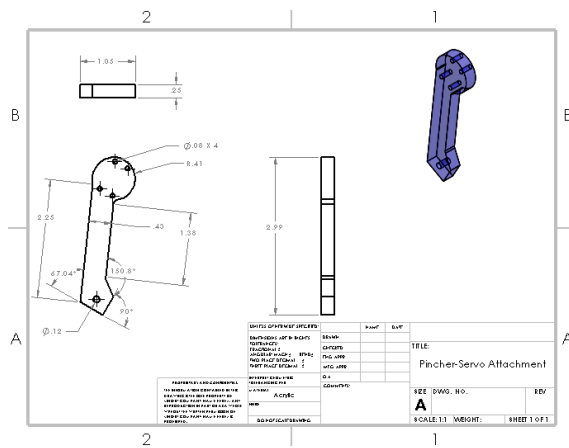
Turning	The new hex rods that replaced the old ones in our gearbox as well as the hex rod used to support our arm mechanism were cut using the lathe.
Milling	The U-Shaped arm stopper, intended to prevent our arm from crashing into the chassis when flipped, was created using the CNC mill.
Laser Cutting	The pincher-servo attachment, left arm slab, right arm slab, and custom-chassis were all cut using the laser cutter.
3D Printing	The custom arm-servo mount, pincher-servo mount, molds for the pinchers, and inserts for both the stationary and moving pinchers were 3D printed. In total we used 9.33 cubic inches of 3D printing filament.
Drilling	We drilled a multitude of components, ranging from holes on the chassis to holes on the aluminum supports that our hex rods would go through.
Molding	We molded both our stationary and moving pincher inserts in Ecoflex 00-50 to increase our coefficient of friction and grip for the pinchers.

Dimensioned Drawings of Non-Standard Parts

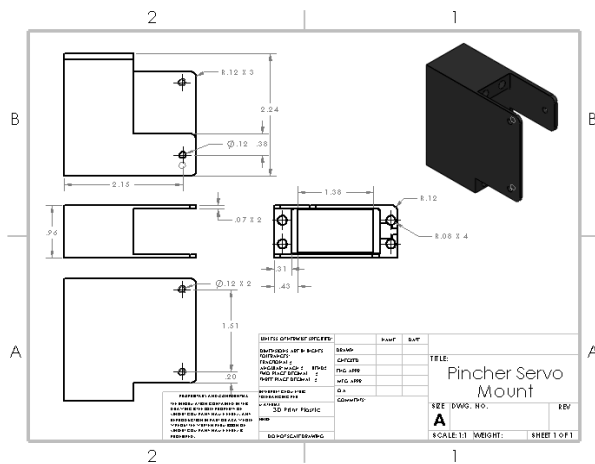
- [Moving Pincher Insert](#)



- Pincher-Servo Attachment

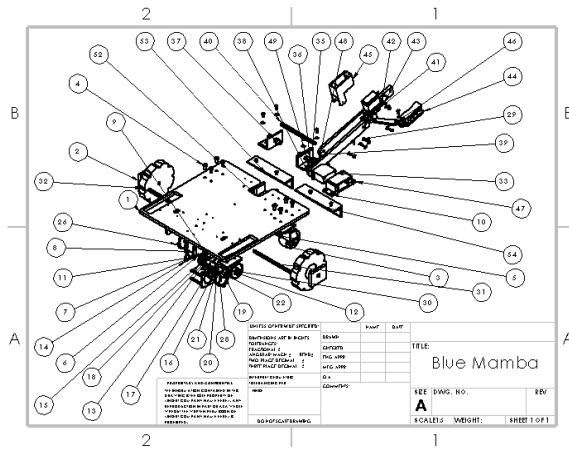


- Pincher Servo Mount

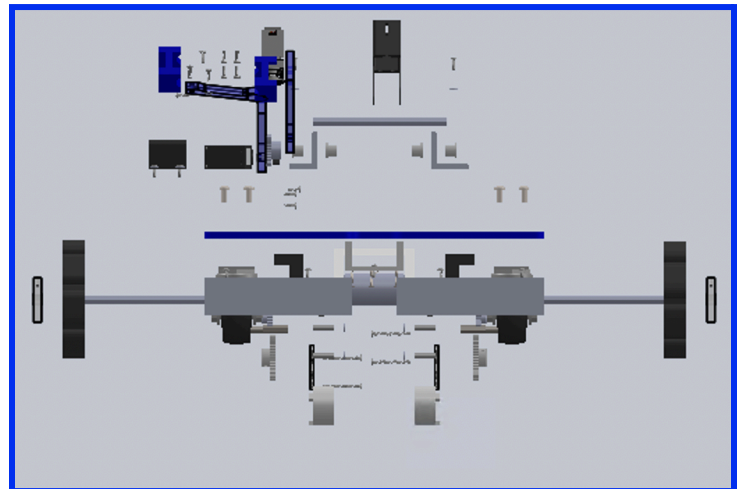
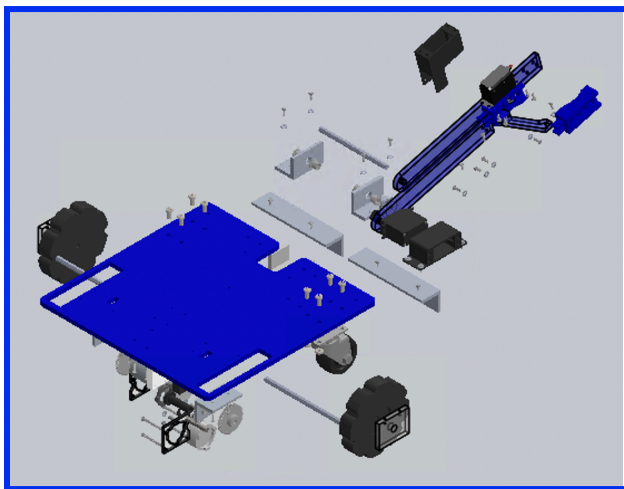


Assembly Drawing of Final Robot

- Corresponding [Bill of Materials](#)



Exploded View of Final Robot



Pictures of Final Robot

